

Dhruv Arora

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CAREER OBJECTIVE

Senior pursuing a double major in Computer Science and Data Science with a Certificate in Economic Analytics at UW–Madison, graduating May 2026. Proficient in data science, AI/ML, and software development, with a strong applied mathematics foundation and hands-on expertise in LLM training and full-stack development. Seeking to leverage a strong foundation in software development, data analytics, and emerging technologies in a dynamic, innovation-driven environment.

SKILLS

- **Programming Languages:** Java, Python, C++, MATLAB, R
- **Web Development:** HTML, CSS, JavaScript, TypeScript, React
- **Databases:** SQL, SQLite, MySQL, JDBC
- **DevOps and Tools:** VS Code, Eclipse, IntelliJ, Git, Docker
- **Machine Learning & AI:** Spark, TensorFlow, Scala, PyTorch
- **Analytical Skills:** Problem Solving, Mathematical Reasoning, Data Analysis, Logical Reasoning
- **Soft Skills:** Focus, Communication, Collaboration, Adaptability

EDUCATION

B.S. in Computer Science and Data Science (Double Major), Certificate in Economic Analytics

University of Wisconsin-Madison, 2022 - 2026 | Cumulative GPA: 3.5

Coursework - Programming (I, II, III), Calculus and Analytic Geometry (1, 2), Elementary Matrix and Linear Algebra, Data Science Modelling (I, II), Data Science Programming II, Computer Engineering, Machine Organization and Programming, Discrete Mathematics, Artificial Intelligence, Algorithms, Human-Computer Interaction, Software Engineering, Database Management Systems

EXPERIENCE

Undergraduate Research Assistant – LLM Training, UW–Madison CDIS, Prof. Shivaram Venkataraman | Aug 2025 – Present

- Verified AI-generated web navigation tasks, tracking sequential steps, parallelization degrees, and URL trajectories to assess retrieval complexity.
- Identified and refined ill-posed or partially answerable prompts, optimizing for clarity and parallel execution in LLM training datasets.
- Generated structured JSON outputs ensuring reproducible, high-quality data for AI model fine-tuning and evaluation.

Data Engineering Intern, Cencora (formerly known as AmerisourceBergen) | June 2025 – August 2025

- Automated SAP spool parsing in Python, standardizing varied log formats and reducing manual effort by 60%.
- Processed hopper files and integrated workflows with Databricks for efficient archival data handling.
- Built a Power BI dashboard to visualize archival data and enable data-driven insights.

Peer Mentor, CS540 – Introduction to Artificial Intelligence, University of Wisconsin–Madison | January 2025 – Present

- Guided students through key AI topics including search, game theory, supervised/unsupervised learning, neural networks, NLP, and reinforcement learning, resulting in improved student understanding and performance
- Led discussion sessions and office hours, collaborating with instructors to enhance course content and reinforce foundational math and logic concepts, which improved student engagement and comprehension.

Undergraduate Student Researcher – ML for Materials Science, UW–Madison College of Engineering | Jan 2024 – May 2024

- Worked under Prof. Dane Morgan to build and evaluate machine learning models using MAST-ML and scikit-learn for engineering datasets, applying cross-validation and hyperparameter tuning, which improved model accuracy and reliability.
- Delivered weekly reports on model improvement, ethical data cleaning, and reproducibility using Python and Google Collab, enhancing team understanding and ensuring data integrity

Full Stack Developer Intern, Wipro Limited, India (Ranked among top 10 IT Services companies in India) | June 2024 – August 2024

- Developed and optimized Java 8 applications (Employee Management System & eShop) with JDBC/MySQL, improving performance, data reliability, and CRUD efficiency by 28%.
- Implemented DAO patterns and secure database interactions using prepared statements and properties files, enhancing maintainability and scalability.

Software Development Intern, Prologic First India Pvt. Ltd., India | June 2023 – July 2023

- Developed a Python chatbot integrating Wit.ai, Snips.ai, and Zapier, improving accuracy by 23% & boosting user engagement.
- Enhanced accessibility with speech-to-text conversion via MediaRecorder API, expanding usability for diverse users.

PROJECTS

- **Puzzle Lab - Crossword Puzzle Web App (Full-Stack Developer)** – Built using React (Next.js, TypeScript) and Quarkus (Java) with JWT authentication, REST APIs, and MySQL backend. Enabled users to create, edit, and play puzzles with personal accounts.
- **NYC Taxi Analysis (Data Engineer)** – Developed a Databricks-based ETL and analytics pipeline for NYC Taxi data, leveraging PySpark, Delta Lake, and MLlib for predictive modelling and visualization.
- **RL - Powered Trading Strategy Optimizer (Machine Learning Engineer)** – Created a DQN-based RL agent to optimize trading strategies, boosting simulated returns by 25% and improving accuracy by 30% using Yahoo Finance data.
- **FashionMNIST Classifier (Deep Learning Engineer)** – Built and trained a neural network using PyTorch, achieving 92% classification accuracy on the FashionMNIST dataset through stochastic gradient descent (SGD) optimization by optimizing prediction confidence via ReLU activations.
- **NOMinate - Smart Meal Planning App (Full-Stack Developer)** – Developed an intelligent meal recommendation app using OpenAI API, Python, and React, leveraging user preferences and real-time feedback to deliver personalized, non-repetitive suggestions.
- **Teeko AI Player (AI Engineer)** – Designed an AI player for Teeko using Minimax with heuristic evaluation, achieving a 40% improvement in strategic decision-making against rule-based opponents.
- **Movie Budget and Revenue Analysis (Data Analyst)** – Analysed the relationship between movie budgets, ratings, and revenue using a dataset of 6820 movies, identifying key success factors and improving budget allocation strategies.

EXTRACURRICULAR ACTIVITIES

- Active member of DotData Club and Software Development Club, UW–Madison – participated in hackathons, coding challenges, and technical workshops.
- Competed in Kaggle competitions and university hackathons, applying AI/ML and software development skills to real-world problems.